

Kaiyuan Liu

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  Scholar
  in Kaiyuan Liu

Education

B.S./M.S. University of Washington, Seattle 2022 ~2027
Combined B.S./M.S. Program in Computer Science
Double Major: Computer Science, Mathematics

- GPA: 3.93/4.0; Honor Math Sequence; Dean's List
- [†]Departmental Honors in Computer Science

Awards

Best Senior Thesis Award, Paul G. Allen School of Computer Science & Engineering 2026.06
ICPC World Final Honor, ICPC Foundation 2024.09
CRA Undergraduate Research Award Honorable Mention, CRA 2025.09
Shenoy Undergraduate Research Fellowship, Simons Foundation 2024 ~2025
CSE Award for Excellence Scholarship, University of Washington 2024 ~2025
ICPC North American Championship 12th Place, ICPC Foundation 2024.05
UW Winter Programming Contest Champion, University of Washington 2023 ~2024
Tinker Research Grant, Thinking Machines Lab 2025.11

Publications

When Agents Slow Down: Understanding LLM Agents' Test-Time Strategies via Elo-per-token Analysis
Kaiyuan Liu, Qiuyang Mang, Bo Peng, Wenhao Chai, Hanchen Li, Shreyas Pimpalgaonkar, Luke Zettlemoyer, Alex Dimakis, Alvin Cheung
arxiv.org/abs/2609.15309 
 arXiv Preprint, 2026

AutoCode: LLMs as Problem Setters for Competitive Programming
 Shang Zhou*, Zihan Zheng*, **Kaiyuan Liu***, *et al.* *Equal contribution
arxiv.org/abs/2510.12803 
 The Fourteenth International Conference on Learning Representations (ICLR 2026)

LiveCodeBench Pro: How Do Olympiad Medalists Judge LLMs in Competitive Programming?
 Zihan Zheng*, Zerui Cheng*, Zeyu Shen*, Shang Zhou*, **Kaiyuan Liu***, *et al.* *Equal contribution
arxiv.org/abs/2506.11928 
 The Thirty-Ninth Annual Conference on Neural Information Processing Systems (NeurIPS 2025)

Harbor Adapters and Harbor-Index: Infrastructure and a Curated Meta-Dataset for Large-Scale Agentic Evaluation
 Lin Shi, Haowei Lin, Zixuan Zhu, ..., **Kaiyuan Liu**, *et al.*
arxiv.org/abs/2609.04298 
 NeurIPS 2026 Under Review

Agents' Last Exam
 Yiyu Sun, Xinyang Han, Weichen Zhang, ..., **Kaiyuan Liu**, ..., Dawn Song
arxiv.org/abs/2606.05405 
 NeurIPS 2026 Under Review

OpenDeepThink: Parallel Reasoning via Bradley-Terry Aggregation

Shang Zhou, Wenhao Chai, **Kaiyuan Liu**, Huanzhi Mao, Qiuyang Mang, Jingbo Shang

arxiv.org/abs/2605.15177 

NeurIPS 2026 Under Review

FrontierSmith: Synthesizing Open-Ended Coding Problems at Scale

Runyuan He, Qiuyang Mang, Shang Zhou, **Kaiyuan Liu**, Hanchen Li, Huanzhi Mao, *et al.*

arxiv.org/abs/2605.14445 

NeurIPS 2026 Under Review

FrontierCS: Evolving Challenges for Evolving Intelligence

Qiuyang Mang*, Wenhao Chai*, Zhifei Li*, Huanzhi Mao*, Shang Zhou*, ..., **Kaiyuan Liu**, *et al.* *Equal contribution

arxiv.org/abs/2512.15699 

Forty-Third International Conference on Machine Learning (ICML 2026)

Evaluating LLM Agents as Human Simulators in Climate Social Dilemmas

Kaiyuan Liu*, Xiaoxuan Hou*, Jiayi Yuan, Natasha Jaques *Equal contribution

[OpenReview Link](#) 

ICLR 2026 MALGAI Workshop *Oral*

Blogpost

Humans Still Beat AI in the Long Horizon: Revisiting Test-Time Scaling in the Agent Era

Qiuyang Mang, **Kaiyuan Liu**, Bo Peng, Shreyas Pimpalgaonkar, Luke Zettlemoyer, Alex Dimakis, Alvin Cheung

frontier-cs.org/blog/humans-still-beat-ai/ 

Frontier-CS Blog, June 16, 2026

Your Next Long-Context Recipe: Open-Ended Problems

Qiuyang Mang, **Kaiyuan Liu**, Zhifei Li, Wenhao Chai, Hanchen Li, Lin Shi, Zhixuan Zhu, Crystal Zhou, Alex Dimakis

frontier-cs.org/blog/harbor/ 

Frontier-CS Blog, May 12, 2026

Experience

Microsoft AI (formerly Microsoft Research – AI Frontiers), Research Intern

Seattle, WA

- Research on self-improvement.

2026.06 ~2026.09

- Advised by Dr. Asli Celikyilmaz.

Social RL Lab, Research Assistant

Seattle, WA

- Working on various projects related to multi-agent, reinforcement learning, and large language models.

2025.01 ~Present

- Mentored by Dr. Natasha Jaques.

Allen Institute, Research Assistant

Seattle, WA

- Applied Deep Learning and Reinforcement Learning to analyze representations from artificial models and neural data from Multi-Armed Bandit mouse foraging tasks. Explored brain representations and dynamics in decision making

2024.09 ~2025.05

VecML, Machine Learning Engineer

Seattle, WA

- Tested and developed vector databases for Retrieval-Augmented Generation.
- Designed and implemented an Memory-Disk Hybrid Architecture for fast Top-K Nearest Neighborhood Search

2024.06 ~2024.09

University of Washington, Teaching Assistant

Seattle, WA

- Assisted in undergraduate algorithm courses for both major (CSE 421) and non-major (CSE 417) students.

2023 ~2024

- Served as leading TA in CSE 421 2024 Winter.

Professional Service

Reviewer, NeurIPS Workshop 2025, NeurIPS 2026, AAAI 2026, AAMAS 2026

ICPC Coach, University of Washington ICPC 2026

Projects

Neural Translator: Predicting Neural Activity from Other Brain Regions

Comp Neuro, NLP

- Developed a neural translator model that predicts neural activity in one brain region from activity patterns in other regions using deep learning and sequence-modeling techniques.

Neural Spike Vectorization with Attention

RL, NLP, Comp Neuro

- Developed an attention-based model for vectorizing brain spike sequences.
- Compared latent representations of RL-trained RNNs with vectorized neural activity from mice performing decision-making tasks using optimal-transport metrics.

Technologies

Languages: Python, PyTorch, JAX, C++, Java, TypeScript, $\text{\LaTeX}$

Databases: MySQL, NoSQL, PostgreSQL, SQLite, Azure

Topics: Algorithm Design, Reinforcement Learning, Machine Learning, NLP, Computational Neuroscience

Languages: English, Chinese (Mandarin)